CHUKWUEMEKA OSMUND AZODO

SENIOR MACHINERY ENGINEER | TURBOMACHINERY RELIABILITY & INTEGRITY | OIL & GAS

B.Eng Mechanical Engineering | ASQ Certified Reliability Engineer (CRE) | COREN R.Engr | ASME, SPE, NSE, ASQ
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BOSIET with CA-EBS (valid to 2029) | Offshore Safety Permit (OSP)

PROFESSIONAL PROFILE

Mechanical engineer with **12 years' experience in turbomachinery** and rotating/packaged equipment reliability, an **ASQ Certified Reliability Engineer (CRE)** and member of several engineering bodies; currently Machinery Engineer for the Oso Gas Hub, the largest offshore gas processing facility in Seplat's JV portfolio. Built on **six years managing machinery efficiency, reliability improvement and technology innovation** for a multi-site gas turbine generating company, driving digital solutions and process innovation that delivered measurable gains in machinery health and operating efficiency. Owns **health-assessment surveillance, troubleshooting and root cause failure analysis** of critical rotating equipment: HP/LP centrifugal compressors, turbo-expanders, industrial gas turbines, cryogenic and API 610 pumps, using vibration, lube-oil and performance-trend analysis via SolarInsight, Seeq and PI XHQ. Reviews equipment and system changes (MOC), stewards maintenance strategy and PSSR, and provides FAT/SAT, commissioning and OEM/vendor assurance for overhauls, brownfield modifications and upgrades. Hands-on and analytical, with a consistent record of converting machinery risk into owned, closed-out actions that protect people, production and asset value.

CORE TECHNICAL COMPETENCIES

Rotating Equipment	Centrifugal gas compressors (HP/LP, recompressors) • Turbo-expander and booster-compressor trains • Industrial gas turbines • Steam turbines • API 610 centrifugal and cryogenic pumps • Diesel engine drivers • Screw and reciprocating air compressors • Dry-gas and mechanical seals, journal/thrust bearings, couplings, lube-oil and seal-gas systems **(OEMs and models listed on page 3)**
Health Assessment & Surveillance	Vibration analysis (ISO 10816/20816, Cat-I) • Lube-oil sampling, elemental & particle-count analysis • Infrared thermography • Videoscope/borescope inspection • Performance monitoring via compressor maps, turbine heat rate and pump curves • Deviation flagging, trending and health reporting to stakeholders
Digital Tools & Programs	SolarInsight • Seeq • PI / PI XHQ historian & dashboards • Honeywell CCC anti-surge and turbomachinery control • SAP PM (notifications, work orders, strategy) • CMMS/EAM data quality • IIoT sensor-to-cloud (Modbus, OPC-UA) • Python/analytics and AI-assisted failure prediction
Reliability & Problem Solving	Root Cause Failure Analysis (RCFA) • RCA (5-Why, Fishbone, Fault Tree) • FMEA / RCM • Crow-AMSAA reliability-growth and Weibull analysis • Bad-actor and MTBF analysis • Criticality ranking • Reliability improvement and defect-elimination programmes
Projects, Lifecycle & Vendor Assurance	Major overhaul planning and execution: gas and steam turbines, compressors, pumps, engines • Brownfield modification and upgrade support • Lifecycle and obsolescence risk management • Execution-ready scopes, datasheets and repair specifications • Vendor/workshop assurance: strip reports, balancing, FAT witnessing • Commissioning and start-up support • PSSR enhancement (off-skid activity capture) and execution
Operations Interface & HSE	Operations/maintenance decision support • Shutdown & turnaround scope optimisation • Risk assessment, HAZID/JSA participation • Permit to Work, LOTO, SIMOPS • OEM and vendor coordination • Multi-discipline technical support and mentoring
Standards	API 610 • API 614 • API 617 • ASME PTC-22 • ISO 10816 / 20816

PROFESSIONAL EXPERIENCE

Machinery Engineer Specialist II, Offshore Asset Engineering — May 2025 – Present

*Amaiden Energy Nigeria Ltd, seconded to **Seplat Energy Producing Nigeria Unlimited** (JV offshore assets: Oso Gas Hub, BRT, UBIT)*

Provides rotating equipment and reliability engineering support across three offshore assets; deployed February 2026 as the dedicated **Machinery Engineer for the Oso Gas Hub** (120 MMSCFD sales gas + NGL export, expanding to 240 MMSCFD), covering top-tier machinery (HP/LP compressors, recompressors, turbo-expanders, cryogenic pumps), utilities (gas turbine generators, firewater pumps, air compressors) and the NGL fractionation train (de-ethaniser, de-propaniser, de-butaniser).

Equipment Health Surveillance & Technical Support

- Run periodic **health assessment surveillance of critical rotating equipment**: review vibration spectra and trends, lube-oil analysis results, compressor performance maps, turbine parameters and pump deliverability curves, and issue analyses and recommendations to operations, maintenance and asset management for every deviation.
- Exploit digital solutions (**SolarInsight, Seeq, PI XHQ**) to trend machinery condition and process parameters, converting raw data into early-warning calls that prevent unplanned trips and deferment.
- Act as the discipline reference point for other engineering and non-engineering functions: process, instrumentation/controls, inspection, operations, planning, procurement and finance, providing rotating-equipment technical input, spares criticality advice and specification support.
- Provide interim **Asset Engineer coverage**, extending beyond machinery into production optimisation, process stabilisation and non-machinery plant systems, ensuring uninterrupted technical support across the facility.

Machinery Problem Resolution, RCFA & Investigations

- **LP gas compressor (130 MMSCFD, 1,440+ psi) severe vibration after bundle change-out:** led the multi-discipline RCA with OEM and in-house teams: mapped contamination paths, ran sectioned borescope inspections of cooler bundles, identified debris-induced rotor unbalance, then directed process piping cleaning, lube-oil flushing and re-inspection. Machine restored to acceptable ISO vibration limits and returned to service.
- **HP gas compression system (120 MMSCFD, 5,500+ psi):** developed and drove the restoration strategy for the facility's most critical train, safeguarding continuity of gas export to the terminal.
- **15-year fleet-wide compressor reliability closeout:** led a data-driven RCFA (2011–2026) using pooled failure-rate and **Crow-AMSAA reliability-growth analysis**, showing the affected compressors failing ~17× more often than a comparable benchmark unit on the same platform. Root-caused the shortfall to a wrong-specification filter element in a housing built for now-discontinued OEM hardware, predating the original 2011 investigation. Engaged the OEM and an alternative supplier on a validated fix, packaged into an eleven-action closeout with named owners and dates.
- **LPA compressor bearing rust contamination:** compared oil signatures across the holding tank, reservoir, pre/post-filtration points and overhead tanks to isolate the source. Fill-and-drain gave no improvement, so escalated to manhole entry and pulse jetting, recovering a thick paste of dirt and rust likely dating to the unit's 1995 commissioning.
- **Depropaniser pump chronic failures (BRT):** RCFA plus an **API 610** review of hydraulic selection, mechanical seal configuration and operating envelope proved the pump was running outside its design point; corrective actions substantially extended run-life and closed out a repeat-failure bad actor.
- **Foundation cracking across multiple pumps and generators (BRT):** led the RCA, established the causal chain and implemented structural reinforcement that restored integrity to the affected rotating equipment foundations.
- **Solar Taurus 60 gas turbine generator failure-to-crank:** supported diagnosis and corrective action with the OEM, restoring power generation availability to the terminal.
- **Firewater diesel engine vibration (UBIT):** resolved a long-standing vibration problem on a safety-critical package and validated performance against pump-curve criteria during deliverability testing.
- Work with operations, maintenance, inspection and electrical teams to resolve **machinery vibration problems across pumps, diesel engines and electric motor drivers** on several offshore platforms and onshore plants: field diagnosis through spectra, phase and alignment checks, corrective action, and post-repair verification against ISO limits.

Risk, Change Review, Projects & Start-up

- Contribute to **risk assessments, HAZID and safety reviews** for machinery interventions, and prepare Lessons Learned under a **PDRR (Prevent, Detect, Respond, Recover)** framework to prevent recurrence.
- Review equipment and system changes (MOC) for their effect on reliability, operability and maintainability; analyse deviations from specification and recommend accept/reject positions.
- Strengthened the facility **Pre-Startup Safety Review (PSSR)** process to capture previously uncaptured off-skid maintenance activities, and now tailor it to each bundle change-out, which typically carries non-standard off-skid scope, verifying safe restart readiness after maintenance and shutdowns.
- Support shutdown and turnaround planning with an emphasis on **optimising machinery scope and duration**, and provide start-up, commissioning and OEM coordination for critical rotating equipment.
- Steward **safe and timely execution of the mechanical scope** for the 120 → 240 MMSCFD expansion: HP 1st-stage wet seal change-out, compressor reconfiguration for export through the 1st stage, decommissioning of the 2nd-stage compressor and associated loops, and gas generator change-out with associated off-skid activities.
- Maintain an appreciable record of identifying and following up on safety hazards under Seplat's **Hazard Identification & Control Scheme (SHICS)**, using the relevant digital reporting tools, with consistent commendation on several occasions.

Manager, Business Innovations, Technology & Efficiency — Jul 2022 – May 2025

First Independent Power Limited (FIPL) / Sahara Power Group, multi-site gas turbine power generation fired on natural and associated gas

- Led **Reliability-Centred Maintenance (RCM)** deployment across the generating fleet, improving availability and MTBF of critical rotating equipment through revised strategies and defect elimination.
- Supervised **gas turbine performance testing to ASME PTC-22**: power output, heat rate and thermal efficiency verification, and used the results to drive degradation-recovery and washing decisions.
- Managed OEM **Factory Acceptance Tests (FAT)** for critical rotating equipment, witnessing performance and mechanical running tests and closing out punch items before shipment.
- Seconded as **Lead Mechanical/Reliability Engineer** for the restart of the Imo River Associated Gas Gathering (AGG) plant, an upstream associated-gas facility, after roughly two years of prolonged downtime: pre-commissioning checks, machinery alignment and verification, first fire and load-up support.
- Designed and deployed **IIoT-based condition monitoring and predictive maintenance** for gas turbines and compressors, enabling real-time machinery health visibility and delivering ~10% operational cost savings.
- Cut compressor wash downtime by **more than 24 hours** per event through procedure and scope optimisation.
- Recognised for safety leadership: **Safety Sentinel** and **Safety Advocate of the Year** awards (2023) for consistent hazard identification and follow-up.

Team Lead, Efficiency & Innovation — Sep 2019 – Jun 2022

First Independent Power Limited (FIPL) / Sahara Power Group

- Led major **gas turbine and compressor overhauls**: planning, clearance verification, rotor and bearing inspection, alignment and re-commissioning, delivering annual savings above NGN 10 million through improved spares management and maintenance planning.

- Chaired cross-functional **RCA investigations** into major turbomachinery failures and drove corrective actions that reduced repeat failures.
- Built real-time plant monitoring dashboards giving operations and maintenance immediate visibility of machinery health, vibration trends and process deviations.
- Coordinated emergency start-ups of critical turbine systems, minimising downtime and restoring generation.

Performance / QA-QC Engineer → Asst. Team Lead, Mechanical Maintenance **Nov 2015 – Sep 2019**

First Independent Power Limited (FIPL) / Sahara Power Group

- Supervised maintenance of high-value turbomachinery and process equipment; delivered quality control during major inspections and overhauls to OEM and industry standards.
- Applied condition monitoring: **vibration analysis, lube-oil analysis, thermography, dissolved gas analysis and motor current signature analysis**, to detect anomalies and prevent unplanned downtime.
- Contributed to review and update of operations and maintenance procedures, strengthening safety protocols and process efficiency.

Graduate Engineer, Mechanical Maintenance **Aug 2014 – Aug 2015**

Sahara Power Group, Egbin Power Plc (1,320 MW thermal station)

- Participated in overhauls of steam turbines, gas turbines, diesel engines and balance-of-plant equipment (pumps, compressors, valves, steam traps, accumulators); gained hands-on exposure to demineralisation, water treatment, desalination and lube-oil/water chemistry systems.

Junior Engineer (Intern), Drillog Petro-Dynamics Ltd **Jun – Dec 2010**

- Maintained directional drilling motors and auto-torque equipment and performed QA/QC inspection of mud motors prior to rig deployment, first exposure to oil & gas service operations.

Mechanical Trainee, Industrial Development Centre, Owerri **2008 & 2009**

- Machining, bench work, precision alignment of rotating components and fault diagnosis using basic root cause techniques.

EDUCATION

M.Eng Mechanical Engineering (Thermo-Fluids), in view — **Rivers State University, Port Harcourt**

B.Eng Mechanical Engineering, Second Class Upper (2:1), 2012 — **Federal University of Technology, Owerri (FUTO)**

ROTATING EQUIPMENT PORTFOLIO: OEMS & MODELS SUPPORTED

Gas Turbines	Siemens SGT-600 • Solar Saturn, Centaur, Taurus and Mars families • GE Frame 5 (MS5001), Frame 9E (MS9001E) combined cycle and GT13E2 • Performance verification to ASME PTC-22; borescope, hot-section and mechanical maintenance support
Centrifugal Compressors	Siemens legacy Dresser-Rand centrifugal compressor family • HP trains to 5,500+ psi (120 MMSCFD) and LP trains to 1,440+ psi (130 MMSCFD) • Recompressors • Anti-surge and turbomachinery control (Honeywell CCC) • Seal-gas, dry-gas seal and lube-oil systems
Expanders & Cryogenic Pumps	Turbo-expander / booster-compressor trains • Cryogenic pumps: Cryostar and Ebara Cryodynamics (Elliott Ebara) • NGL fractionation: de-ethaniser, de-propaniser, de-butaniser • Sales gas (C1/C2) and NGL (C3+) export systems
Pumps, Drivers & Auxiliaries	API 610 centrifugal process pumps • Firewater diesel engine packages and deliverability testing • Steam turbines (Egbin 1,320 MW station) • Screw and reciprocating air compressors • Electric motor drivers and gearboxes • Lube-oil and seal-oil consoles, run-down tanks, coolers, filtration • Bearings, couplings, mechanical seals

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- ASQ Certified Reliability Engineer (CRE), 2024
- Vibration Analysis Category I, Istec Academy, 2024
- Reliability Engineering Principles, Beina M&R, 2024
- Gas Turbine Performance, Entropy-Tech, 2021
- Gas Turbine Mechanical Maintenance, Masaood John Brown, 2018
- BOSIET with CA-EBS (2029) & Offshore Safety Permit (OSP)
- HSE Level I, II & III (OSHA), HSETrain Intl., 2020
- QHSE, Risk & Sustainability Mgmt, IE Safetainability, 2024
- AI Advanced Applied Programme, Digital Regenesys, 2026
- IBM AI Engineering, 2023 & Generative AI and Agents, Johns Hopkins, 2026

PROFESSIONAL MEMBERSHIPS & RECOGNITION

- American Society for Quality (ASQ), 2024
- Society of Petroleum Engineers (SPE), 2022
- American Society of Mechanical Engineers (ASME), 2021
- COREN Registered Engineer, 2019
- Nigerian Society of Engineers (NSE), 2017
- Safety Advocate of the Year, FIPL, 2023
- Safety Sentinel Award, FIPL Trans-Amadi, 2023
- 1st Runner-Up, Sahara Group Innovation Hackathon, 2020
- Chairman's Recognition for Innovation, Sahara Group, 2019

References: available on request.